Methods: Source-Owned Token Injection and Conditional IMRAD Assembly

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The method is conditional section hydration. Each slot combines the seed, slot name, category, ordinal, and full category list into a SHA-256 digest; the digest indexes the configured category. Including the full category list in the digest input means that a lexicon edit can change the plan in a deterministic and reviewable way instead of silently preserving stale output.

The deterministic digest recipe is deliberately narrow. It does not sample from ambient prose, project history, or renderer state; it uses only the committed seed, the slot declaration, the category name, the ordinal for repeated slots, and the ordered category inventory. A fork can therefore explain a changed token choice as a changed seed, slot, or lexicon row rather than as an opaque generation event.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

The first review scenario is declared before generation. The project names its scope as a local exemplar, records enabled sections, treats any publication identifiers as metadata rather than method evidence, and treats the copy stage as a review handoff. That ordering matters because a reader should know the allowed claim boundary before inspecting fluent generated text.

The governing constraint is publication claims stay local until release. The source manuscript is intentionally sparse: it contains section titles and large-grain placeholders, not generated claims. The project script first validates the `madlib:` block, expands slots into a `TokenPlan`, builds section bodies, writes artifact JSON, emits a figure registry, and only then writes hydrated Markdown under `output/manuscript/`.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

The configured method moves are validate config before composition, declare review scenario and method invariants, separate explicit YAML fields from loader defaults, govern lexicon categories as reviewable data, expand slots through seeded digest selection, allocate slot choices to enabled manuscript sections, compose conditional section bodies, assemble method evidence tables and visual audit figures, align generated claims with the claim ledger, emit evidence artifacts before rendering, verify that no unresolved placeholders remain, assemble a reviewer packet from regenerated artifacts, and preserve human review before publication claims. The protocol sequence is Ingest declared manuscript schema produces `MadlibConfig`; Declare review scenario produces `review scenario`; Track field origin produces `explicit/default path inventory`; Govern lexicon categories

produces `validated lexicon`; Construct digest selection material produces `digest input records`; Record selection invariants produces `selection invariant set`; Expand slot declarations produces `TokenPlan`; Apply section conditions produces `enabled section set`; Compose conditional IMRAD bodies produces `section variables`; Assemble evidence tables produces `Markdown evidence tables`; Align claims with evidence ledger produces `claim-aligned evidence surface`; Generate visual audit surface produces `registered figure set`; Emit machine-readable artifacts produces `output/data, output/reports, and output/figures`; Hydrate manuscript shell produces `output/manuscript`; Render and validate deliverables produces `validated project output`; Assemble reviewer packet produces `review packet`; Copy review

surface produces `copied publication-review bundle`; Document fork migration produces `fork migration notes`. These steps make the method auditable from three directions: tests inspect the Python behavior, generated artifacts expose the token plan, and manuscript validation confirms that no unresolved placeholder survives into rendered outputs.

The config-origin inventory currently separates 126 explicit YAML path(s) from 11 loader-defaulted path(s). Treating origin as method evidence prevents a rendered field from looking equally authored when it was actually inherited from the loader. The same inventory drives configured-field tables and figures, so reviewers can inspect which schema blocks were intentionally set before judging the generated prose.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

Method invariants are reviewed as their own artifact. Token choices are allowed to change when the seed, slot name, category, ordinal, or ordered category inventory changes; they are not allowed to change because PDF rendering, HTML rendering, file-copy order, or hand-edited output changed. This separates generation logic from presentation logic.

Lexicon governance is handled as data governance. Required categories must be nonempty, optional categories remain project-owned when declared, and the selected 40 token choice(s) stay bound to 10 configured category list(s). The slot-to-section allocation is abstract: 3, authoring_contract: 2, configuration: 1, discussion: 3, evaluation: 5, introduction: 8, limitations: 3, methods: 6, reproducibility: 2, results: 5, scope: 2, which lets the Methods, Results, and provenance tables state where each lexical decision enters the manuscript.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

Conditional section generation is handled before prose assembly. A disabled section does not vanish and does not borrow claims from an enabled section; it resolves to an explicit statement that names the controlling `madlib.section_conditions` key. That behavior keeps negative or excluded material visible to reviewers.

Evidence tables and visual audit figures are generated from the same config and `TokenPlan`. Enabled visualizations are configured_field_matrix, section_configuration_heatmap, field_origin_summary, token_injection_flow, section_token_allocation, provenance_trace_map, quality_gate_matrix; they summarize field origin, token density, injection flow, section allocation, provenance, and quality gates without adding independent claims. Figure registration is therefore a method step: every manuscript image has to be written, registered, and validated as part of the reproducible render path.

Claim-ledger alignment is part of composition. The contribution table can make local method claims, but publication, empirical, reader-quality, and domain-specific claims must either point to evidence or remain explicit non-claims. This is why the claim ledger, audit rules, limitations, and authoring contract are generated beside the Methods rather than written after the fact.

Evaluation is part of the method rather than an afterthought. The config declares quality probes (Method row completeness, Field-origin visibility, Placeholder survival, Provenance completeness, Section-switch observability, Figure registry coverage, Method-figure alignment, Evidence cleanliness, Fork readiness, Copied-output parity, Digest invariant review, Claim-ledger alignment, Review packet completeness, Fork migration sufficiency) and failure modes (Unresolved placeholder, Overclaimed generated prose, Config-source drift, Figure

provenance gap, Domain misuse, Method row drift, Field-origin opacity, Visual-method mismatch, Fork without validators, Digest invariant drift, Claim ledger omission, Review packet incompleteness, Fork migration ambiguity); the source turns them into tables and validation checks the rendered surface. That means methods, results, evaluation, and limitations all share one source-owned schema instead of drifting as independent prose.

The method is organized around design principles: Configuration owns prose choices, Method surface is config-owned, Token choice is deterministic, Field origin is evidence, Sections are conditional but visible, Visual audit follows data, Generated output is disposable, Claim boundaries travel with prose, Forks must add validators, Invariants precede rendering, Diffs are review objects, Review packet is a method artifact, Fork migration is part of the method. These

principles prevent the Mad Lib surface from becoming a hidden authoring channel. They require the visible manuscript to stay downstream of declared inputs, the generated outputs to remain disposable, and the audit surface to be broad enough for a reviewer to reconstruct how a sentence reached the PDF.

The operational phases are Schema intake maps `manuscript/config.yaml` to `MadlibConfig`; Scenario declaration maps `MadlibConfig` to `review scenario`; Field-origin inventory maps `MadlibConfig and raw YAML keys` to `configured_field_inventory.json`; Lexicon validation maps `madlib.lexicon and madlib.slots` to `validated slot inventory`; Digest token planning maps `MadlibConfig` to `TokenPlan`; Invariant review maps `TokenPlan and method protocol` to `selection invariant set`; Slot-to-section allocation maps

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

TokenPlan to section token counts; Section composition maps TokenPlan and narrative moves to manuscript variable map; Evidence table assembly maps MadlibConfig and TokenPlan to manuscript_variables.json; Claim-ledger alignment maps MadlibConfig, generated prose, and data/claim_ledger.yaml to claim-aligned evidence surface; Visualization emission maps MadlibConfig, TokenPlan, and configured-field inventory to output/figures and figure_registry.json; Artifact emission maps MadlibConfig and TokenPlan to output/data, output/reports, and output/figures; Manuscript hydration maps source manuscript shells and manuscript_variables.json to hydrated Markdown manuscript; Render maps output/manuscript to output/pdf, output/web, and output/slides; Validate and copy

maps `project output directories` to `output/templates/template_madlib`; Review packet assembly maps `validated project output and copy statistics` to `review packet`; Fork contract documentation maps `source docs, authoring contract, and claim ledger` to `fork migration notes`. Each phase has an explicit input, transformation, output, and guard. This makes the pipeline explainable at manuscript scale: a reader can follow the path from YAML declarations to token choices, from token choices to section bodies, from section bodies to rendered artifacts, and from rendered artifacts to validation reports.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

The reviewer packet is also a method artifact. The handoff surface is hydrated Markdown, combined PDF, web output, slides, figures, data JSON, reports, validation results, and copy statistics; a PDF alone is insufficient because it cannot show the token inventory, field-origin inventory, figure registry, validation report, or copied-output statistics. The declared authoring obligations are Review generated claims, Review config diffs, Extend claim evidence, Add domain validators, Rerun the full project path, Review method invariants, Assemble reviewer packet, Write fork migration notes, which convert that packet into review work a human can actually perform.

The claim-boundary contract is also generated. The audit-rule list contains 12 rule(s), and the contribution table binds each local claim to a non-publication boundary. The final copy stage is a human-review handoff, not proof that the Mad Lib surface is empirically valid or ready for a standalone release.

Fork migration closes the method. A downstream project should update config rows, source-owned composition, validators, claim-ledger entries, and documentation before replacing exemplar vocabulary with domain claims. Without that migration work, the fork has only changed words, not the evidential status of the generated manuscript.

Methods: Source-Owned Token Injection and Conditional IMRAD Assembly I

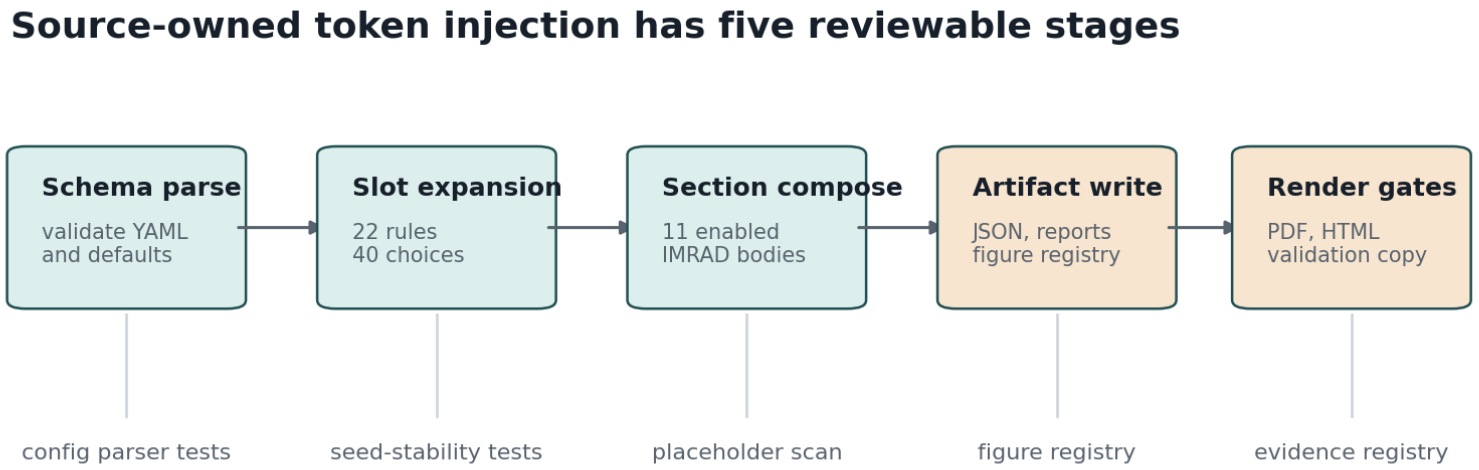


Figure 1: Deterministic token-injection flow

Design Principles I

Design Principles II

Principle	Rationale	Manuscript effect
Configuration owns prose choices	Reviewers can inspect the declared language surface before generation.	Large-grain manuscript variables are generated from YAML and project source.
Method surface is config-owned	A fork should change method protocol rows before changing generated Methods prose.	The Methods tables and body summarize method_protocol and pipeline_phases from YAML.
Token choice is deterministic	A fixed seed and lexicon must produce the same injection plan across reruns.	Token inventory rows can be regenerated and diffed.
Field origin is evidence	A generated manuscript should distinguish authored YAML fields from loader defaults.	Configured-field tables and figures report explicit and defaulted paths.
Sections are conditional but visible	Disabled material should be auditable rather than silently absent.	Every disabled section resolves to an explicit disabled-section body.
Visual audit follows data	Figures should explain the generated method without becoming decorative claims.	Cover, flow, allocation, provenance, gate, and field-origin figures are regenerated from artifacts.
Generated output is disposable	The durable artifact is the regeneration contract, not hand-edited output files.	Output Markdown, PDF, HTML, and slides are rebuilt from source inputs.
Claim boundaries travel with prose	Generated text can otherwise imply validation that no artifact supports.	Contribution, limitation, failure-mode, and authoring tables carry boundary text.
Forks must add validators	Domain claims need domain evidence beyond this exemplar's generic regeneration gates.	Scope, limitations, and authoring contract require validators before domain claims.
Invariants precede rendering	Readers need to know which inputs are allowed to alter generated tokens before they inspect output.	Methods names the digest inputs and excludes renderer state from token choice.
Diffs are review objects	Config, token inventory, manuscript variables, figures, and copied outputs should be diffable review surfaces.	Methods and reproducibility text describe review packet assembly after regeneration.
Review packet is a method artifact	A PDF alone is insufficient evidence for a generated-method exemplar.	Authoring contract and validation sections treat data, reports, figures, and logs as part of review.
Fork migration is part of the method	A public exemplar should teach authors what must change before domain use.	Standalone notes, authoring obligations, and claim-ledger boundaries name fork responsibilities.

Operational Phases I

Operational Phases II

Phase	Input	Transformation	Output	Guard
Schema intake	`manuscript/config.yaml`	Load paper metadata and validate the madlib schema before generation.	`MadlibConfig`	config parser tests
Scenario declaration	`MadlibConfig`	Summarize local scope, enabled sections, claim boundaries, and review handoff expectations.	`review scenario`	method protocol and contribution table tests
Field-origin inventory	`MadlibConfig and raw YAML keys`	Classify supported paths as explicit or defaulted.	`configured_field_inventory.json`	configured-field inventory tests
Lexicon validation	`madlib.lexicon and madlib.slots`	Reject empty required categories and slot references to missing categories.	`validated slot inventory`	malformed-config tests
Digest token planning	`MadlibConfig`	Hash seed, slot name, category, ordinal, and category inventory.	`TokenPlan`	seed-stability tests
Invariant review	`TokenPlan and method protocol`	Confirm allowed token-choice inputs are documented and isolated from renderer state.	`selection invariant set`	token determinism and Methods prose tests
Slot-to-section allocation	`TokenPlan`	Assign each selected token to its configured manuscript section.	`section token counts`	provenance and allocation tests
Section composition	`TokenPlan and narrative moves`	Build conditional section bodies, titles, and evidence tables.	`manuscript variable map`	placeholder-coverage tests
Evidence table assembly	`MadlibConfig and TokenPlan`	Render protocol, phase, audit, token, section, provenance, and configured-field tables.	`manuscript_variables.json`	composition table tests
Claim-ledger alignment	`MadlibConfig, generated prose, and data/clai`	Check local claims and non-claims against	`claim-aligned evidence surface`	claim ledger and evidence registry review

Protocol Steps I

Protocol Steps II

Step	Action	Evidence	Output
Ingest declared manuscript schema	Parse paper metadata and the madlib block from manuscript/config.yaml before any prose or figures are composed.	Config validation tests and MadlibConfig construction from the committed YAML.	`MadlibConfig`
Declare review scenario	Name the manuscript scope, local claim boundary, enabled sections, and intended reviewer handoff before token generation.	section_plan.json, contribution table, and authoring contract rows.	`review scenario`
Track field origin	Record every supported madlib path as explicit when it appears in YAML or defaulted when the loader supplies it.	configured_field_inventory.json and configured-field origin tests.	`explicit/default path invent ory`
Govern lexicon categories	Reject empty required categories, preserve project-owned optional categories, and treat every lexical list as source data.	Malformed-config tests and lexicon rows in token_inventory.json.	`validated lexicon`
Construct digest selection material	Combine seed, slot name, category, ordinal, and the full category inventory into a deterministic digest input.	Seed-stability and category-sensitivity tests.	`digest input records`
Record selection invariants	State the invariant inputs that are allowed to change token choices and the renderer state that is not allowed to affect them.	Token determinism tests and Methods digest prose.	`selection invariant set`
Expand slot declarations	Resolve every slot count into one or more token choices and assign each selected value to its configured manuscript section.	TokenPlan construction, provenance trace, and section allocation figure.	`TokenPlan`
Apply section conditions	Evaluate section switches before prose assembly so disabled	Section-condition tests and output/data/section_plan.json	`enabled section set`

Section Plan I

Section Plan II

Section	Render title	Enabled	Token choices	Narrative moves
abstract	Abstract	True	3	state the manuscript-generation problem, name the deterministic intervention, summarize the audit surface
introduction	Introduction: Lexicon as Data and Manuscript as Build Artifact	True	8	separate playful Mad Lib syntax from research claims, identify drift between prose and source data, frame configuration as an inspectable dataset, position conditional prose as a reproducibility problem
methods	Methods: Source-Owned Token Injection and Conditional IMRAD Assembly	True	6	validate config before composition, declare review scenario and method invariants, separate explicit YAML fields from loader defaults, govern lexicon categories as reviewable data, expand slots through seeded digest selection, allocate slot choices to enabled manuscript sections, compose conditional

Audit Rules I

Audit Rules I

Rule	Enforcement surface
R1	Every manuscript placeholder must be generated by source code.
R2	Every generated token choice must carry category and config-key provenance.
R3	Every method protocol row must identify an action, evidence artifact, and output.
R4	Every pipeline phase must identify an input, transformation, output, and guard.
R5	Every visible configured field must be classified as explicit or defaulted.
R6	Every disabled section must resolve to an explicit disabled-section body.
R7	Every figure reference must be backed by a generated figure registry entry.
R8	Every fork that adds domain claims must add domain validators and claim-ledger evidence.
R9	Every publication claim must stay local unless a live DOI or release exists.
R10	Every token-selection explanation must name only seed, slot, category, ordinal, and ordered category inventory as digest inputs.
R11	Every review handoff must include generated data, reports, figures, validation results, and copy statistics alongside PDF or HTML.
R12	Every fork migration note must name config, source, test, validator, pipeline, and claim-ledger obligations.